



A comparative study of machine learning methods for bio-oil yield prediction – A genetic algorithm-based features selection

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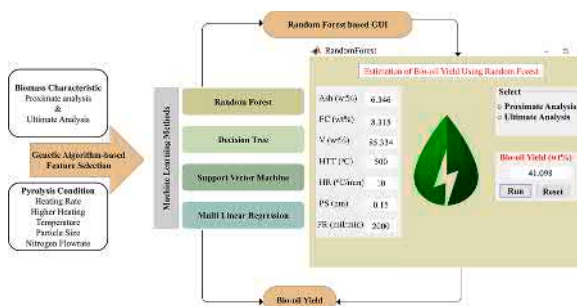
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HIGHLIGHTS

- A genetic algorithm-based approach was used for feature selection.
- Random forest outperformed all other ML models in predicting bio-oil yield.
- Analysis of Partial Dependence Plot showed inside details for pyrolysis process.
- A Graphical User Interface for predicting bio-oil yield was developed.

GRAPHICAL ABSTRACT



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ABSTRACT

A novel genetic algorithm-based feature selection approach is incorporated and based on these features, four different ML methods were investigated. According to the findings, ML models could reliably predict bio-oil yield. The results showed that Random forest (RF) is preferred for bio-oil yield prediction ($R^2 \sim 0.98$) and highly recommended when dealing with the complex correlation between variables and target. Multi-Linear regression model showed relatively poor generalization performance ($R^2 \sim 0.75$). The partial dependence analysis was done for ML models to show the influence of each input variable on the target variable. Lastly, an easy-to-use software package was developed based on the RF model for the prediction of bio-oil yield. The current study offered new insights into the pyrolysis process of biomass and to improve bio-oil yield. It is an attempt to reduce the time-consuming and expensive experimental work for estimating the bio-oil yield of biomass during pyrolysis.

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Nomenclature

ML	Machine Learning
RF	Random Forest
MLR	Multi Linear Regression
SVM	Support Vector Machine
DT	Decision Tree
ANN	Artificial Neural Network
LSTM	Long Short-term Memory
LR	Linear Regression
GA	Genetic Algorithm
FC	Fixed Carbon
V	Volatile Matter
HTT	High Treatment Temperature
HR	Heating Rate
PS	Particle Size
FR	Flow Rate (Nitrogen)
R ²	Coefficient of Determination
RMSE	Root Mean Square Error
MSE	Mean Square Error
OOB	Out-of-the-bag
IB	In-bag

1. Introduction

Fossil fuel-based energy consumption is on the rise due to many emerging economies and an increase in the human population. This leads to depleting fossil fuel reserves in many countries, an increase in greenhouse gas emissions, and a rise in the market price of fuels (Gustafson et al., 2020; Plantinga & Scholtens, 2021). Renewable energy sources could be the potential alternatives to cope with the above-mentioned problems and to reduce fossil fuel geographical dependency (Arachchige et al., 2021). Various renewable energy sources, including solar energy, wind energy, hydro-energy, biomass-energy, and geothermal energy are potential alternatives (Jurasz et al., 2020; Sinsal et al., 2020). Among these, biomass energy, or “Bioenergy” is the most promising and sustainable source of energy that can be replaced with fossil fuels for both energy and chemical production. Biomass is an organic material mainly derived from plants, including agriculture and forestry residue, municipal solid wastes, food waste, and sewage sludge (Sirohi et al., 2020; Wang et al., 2020). It can be converted into solid, liquid, and gaseous products through biochemical and thermochemical conversion techniques. In the past decades, significant progress has been made and scientists were able to devise thermochemical processes that are low cost and offer relatively high conversion performance with simple pretreatment (Ong et al., 2020; Parvez et al., 2020).

Pyrolysis is one of the thermochemical conversion processes in which biomass feedstock is decomposed to produce bio-oil, biochar, and non-condensable gas in the absence of air. The liquid product, bio-oil or pyrolysis oil, is a dark brown viscous liquid generally composed of 350 highly oxygenated compounds (Lu et al., 2020; Ryu et al., 2020). The quantity or yield of bio-oil mainly depends upon the biomass feedstock composition and pyrolysis conditions (Fan et al., 2021). The biomass feedstock composition is usually characterized by proximate and ultimate analysis. The proximate analysis is used to determine the fixed carbon, moisture, volatile, and ash contents of biomass while the ultimate analysis is used to determine the elemental composition of biomass including carbon, hydrogen, nitrogen, and oxygen contents. The pyrolysis process is affected by various variables such as pyrolysis temperature, heating rate, biomass particle size, and residence time (X. Bai et al., 2020; Riva et al., 2021). The influence of biomass feedstock composition and pyrolysis conditions on the production of bio-oil has been investigated in several studies. For instance, (Gholizadeh et al., 2019)

investigated bio-oil production from 20 different biomass feedstock and concluded that the average bio-oil yield was higher from the woody biomass (52 wet wt.%) compared to herbaceous biomass (38 wet wt.%). Similarly, (Sarkar & Wang, 2020) performed slow-pyrolysis of waste coconut shell to investigate the effect of temperature on bio-oil yield and found that the highest yield of bio-oil (48.7 wt%) was recorded at 600 °C. In another study, (Hao et al., 2021) concluded that the bio-oil derived from the mixture of rice straw (RS) and *Ulva prolifera* macroalgae (UPM) yielded maximum bio-oil (46.68 wt%) at 500 °C. Effect of reaction temperature on bio-oil yield was investigated by (Hanif et al., 2016) and they discovered that an intermediate temperature of 300–350 °C maximized the bio-oil yield (48 wt%) produced from algae biomass.

However, the conventional strategies for investigating the bio-oil yield and its correlation with its influencing factors (biomass composition and pyrolysis conditions) involve detailed experimentation, which is labor-intensive, time-consuming, and expensive to execute. Therefore, it is essential to examine the behavior of biomass pyrolysis with combined considerations of feedstock composition and pyrolysis process conditions using machine learning, deep learning, and data mining techniques, to effectively analyze their cumulative influence on the bio-oil yield.

With the emergence of artificial intelligence (AI), many state-of-the-art and novel methods have been incorporated into traditional methods that work well with both linear and nonlinear problems (Quaranta et al., 2020; Tang et al., 2020). Machine learning (ML), a subset of AI, techniques such as support vector machine (SVM), random forest (RF), multilinear regression (MLR), and decision tree (DT) have been successfully implemented in pyrolysis of biomass and showed remarkable performance as compared to the conventional strategies due to their strong ability to predict the outcome (Chen et al., 2018; Mutlu & Yucel, 2018).

Linear regression analysis is widely used on datasets with a linear relationship between input and target variables. For instance, (Hussain & Mustafa, 2016) developed a linear regression model in excel for bio-oil production from the biomass via fast pyrolysis by correlation of reaction temperature, retention time, and biomass composition with the bio-oil yield. The results showed that the coefficients of determination for different regression models were from 0.81 to 0.99. In another study, a linear regression method was developed (Li et al., 2017) to investigate the correlation between 20 different samples of biomass feedstock and the distribution of the bio-oil compound. It was concluded by the authors that there is a strong correlation between bio-oil compounds and biomass composition. Woody biomass yielded more phenols, while straw yielded more ketones, shell biomass yielded more furans, and algae biomass yielded more fatty acids. However, the linear regression-based models only consider the linear correlation between the variables and are not acceptable for the complex reactions that are taking place and need nonlinear correlations. Moreover, these models were usually constructed with a few experimental points in terms of many influential variables, reducing the model's reliability and generalization. Therefore, it is crucial to conduct a comparative study of different predictive ML based models for nonlinear datasets for pyrolysis of biomass with overall consideration of biomass composition and pyrolysis conditions.

In this study, various ML models including MLR, RF, SVM, and DT were developed for the prediction of the bio-oil yield using biomass composition (ultimate and proximate analysis) and pyrolysis conditions (heating rate, highest pyrolysis temperature, particle size, and nitrogen flow rate). A key novelty of this work is that ML models were developed with good prediction performance, using genetic algorithm-based features selection approach rather than traditional approaches such as filter and wrapper methods. In the filter method, features are selected based on the relationship with the target and input variables instead of cross-validation performance. This approach may mislead the algorithm for feature selection. Whereas, in the wrapper method, the actual ML model is trained on each set of features to select the most relevant features.

Which is relatively time-consuming and may lead to overfitting due to the training of ML models based on different features. While the GA-based feature selection method used in this research is a novel and advanced optimization method that employs a natural approach to problem-solving. It is less time-consuming and improves the results based on the cross-validation performance. Lastly, an easy-to-use software package was developed based on the proposed random forest model for the prediction of bio-oil yield. The predictive accuracy of ML models and new perspectives behind the models offered a thorough understanding of biomass pyrolysis and technical guidelines for improving bio-oil production.

2. Materials and methods

2.1. Data collection and Pre-processing

A systematic literature review was conducted to collect the data related to the experimental studies of bio-oil yield from biomass. The articles published on the topic were searched in the citation databases of Google Scholar, ScienceDirect, and Web of Science using different keywords such as bio-oil yield, pyrolysis product, machine learning, lignocellulosic biomass, herbaceous plants, food waste, and algae, etc. After the literature retrieval process, the relevant articles were selected followed by data extraction and compilation for developing and testing the machine learning-based predictive models. The strategies followed for the data collection are described below.

- A total of 263 data points on bio-oil yield were collected from 20 published articles with a total of 15 kinds of lignocellulosic biomass, 12 kinds of herbaceous plant, and 6 kinds of algae-based feedstock.
- Each sample collected was taken from inert environment (oxygen-free environment) experiments based on biomass composition and pyrolysis conditions in the fixed-bed environment.

- The data points were gathered from the tables, extracted from the figures, and additional materials in the final set of relevant literature.
- Data points on biomass ultimate analysis as Carbon (C), Hydrogen (H), Nitrogen (N), Oxygen (O) contents, and proximate analysis as Fixed carbon (FC), Volatile matter (V), and Ash results were collected.
- Pyrolysis conditions such as Heating rate (HR), Highest Treatment Temperature (HTT), Particle size (PS), and Nitrogen flow rate (FR) were taken as input parameters.
- The bio-oil yield was expressed on a dry-ash basis in this study.
- Finally, an easy-to-use Graphical User Interface (GUI) has been developed using the MATLAB Guide Toolbox. The developed GUI was based on the proposed proximate and ultimate models of RF. In the interface of the software, the user will only need to put the values of the selected features in the software and will get the predicted value of bio-oil yield.

The schematic view of the structure of the research methodology includes problem statement, data handling, developing machine learning models, and an easy-to-use graphical user interface for bio-oil yield prediction is shown in Fig. 2.

A boxplot was employed for data distribution showing the quartile, mean, median, minimum, and maximum values. The description of the dataset for bio-oil is shown in Fig. 3. The triangular-shaped box in the figure represents the minimum and maximum values. The solid line shows the mid-value of the sample dataset, while the top and bottom of the box denote 25 percent and 75 percent of the sample distributions, respectively. For the dataset of bio-oil yield, the values of proximate analysis (FC, V, and Ash) were in the ranges of 3.22%–31.13% (FC) 51.45%–87.04% (V), and 0.24%–17.33% (Ash), respectively while the ultimate analysis (C, H, N, and O) were between 38.5% and 61%, 4%–9%, 0.25%–9.30%, and 28%–54%, respectively. Similarly, the pyrolysis conditions (HTT, HR, PS, and FR) for the bio-oil yield dataset were

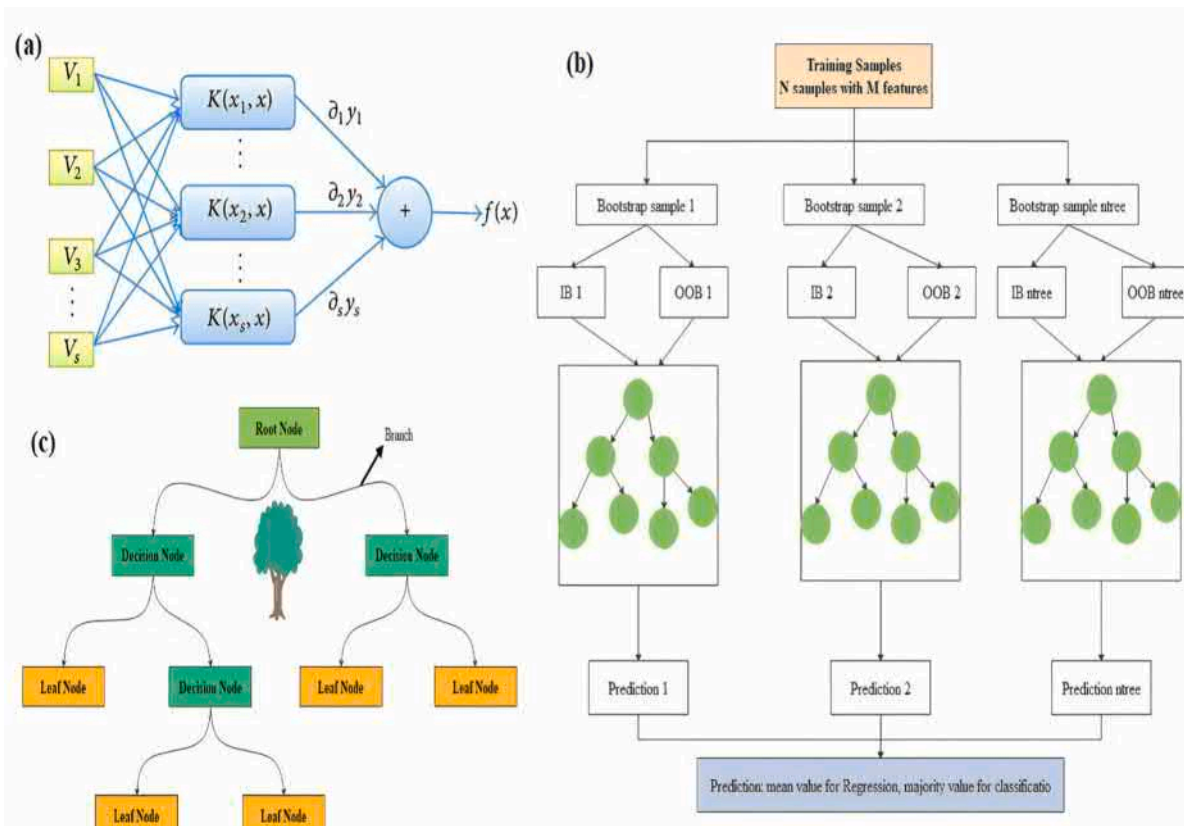


Fig. 1. Schematics of Machine Learning Algorithms: (a) Support Vector Machine, (b) Random Forest, (c) Decision Tree.

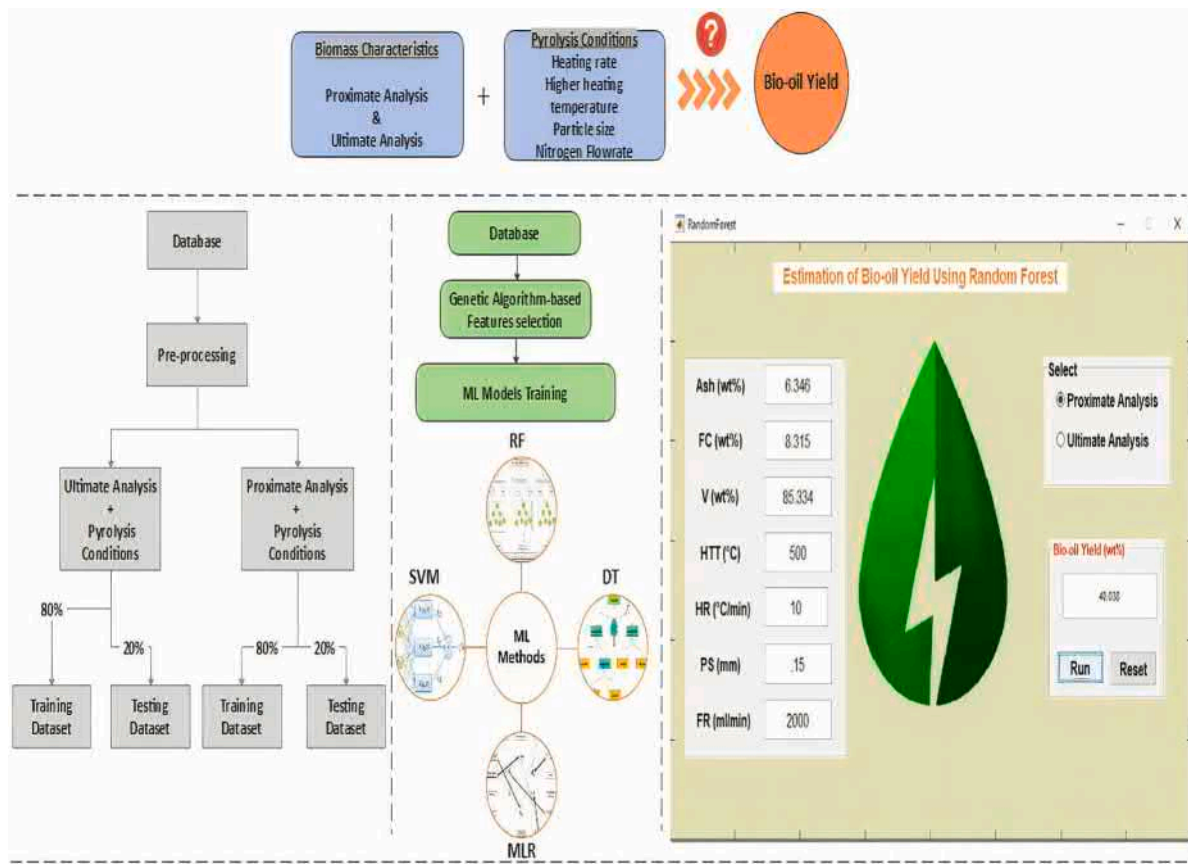


Fig. 2. The schematic view of the research methodology, stating the problem, data handling, developing machine learning models, and easy-to-use graphical user interface for bio-oil yield prediction.

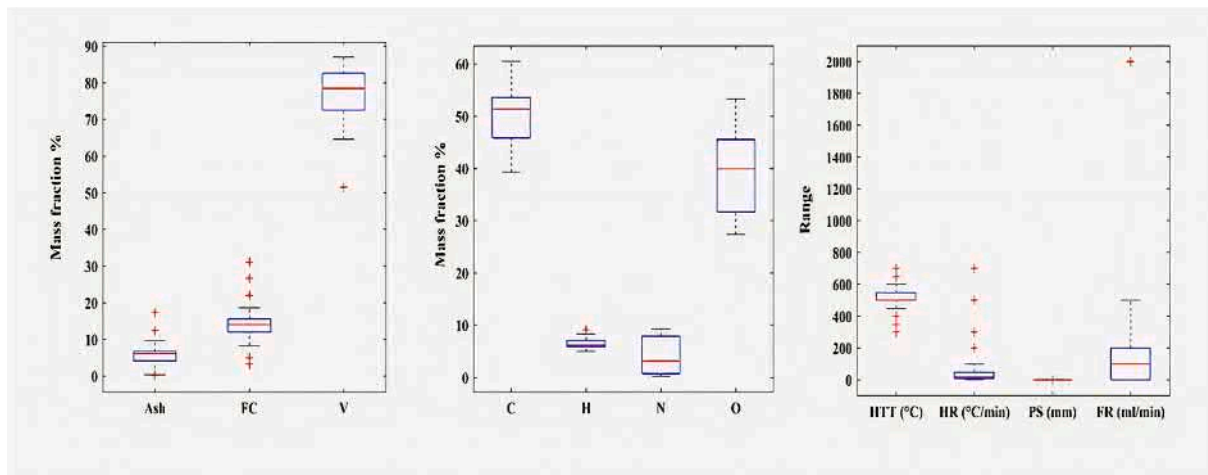


Fig. 3. Box plot of the dataset.

between 250 and 750 °C (HTT), 1.5–760 °C/min (HR), 0.20–2 mm (PS), and 0–1990 mL/min (FR), respectively.

2.2. Performance evaluation criteria

The performance evaluation of each ML model was examined using the following two criteria i) Coefficient of determination (R^2) and ii) Root-mean-squared error (RMSE). The equations for R^2 and RMSE are given as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_i^{\text{exp}} - Y_i)^2}{\sum_{i=1}^n (Y_i^{\text{exp}} - Y_{\text{avg}})^2} \quad (5)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i^{\text{exp}} - Y_i)^2} \quad (6)$$

Where n represents the number of test samples, Y_i^{exp} denotes the experimental value, and Y_i represents the predicted value. $Y_{\text{avg}}^{\text{exp}}$ denotes the

mean value of Y_i^{exp} and Y_i , respectively.

2.3. Machine learning methods

In this work, various machine learning (ML) methods namely, RF, MLR, SVM, and DT are considered for the prediction of bio-oil yield. These methods are briefly described below. Fig. 1 shows the schematics of the machine learning algorithms.

Random Forest (RF): Random forest is an ensemble ML method that can be used to solve classification and regression problems. The key component in the RF is the decision tree. Comprehension of random forest, therefore, required the knowledge of the decision tree (DT). The following are the main steps to develop the RF model. The training data set is divided by the bootstrap strategy into several randomly generated subsets. The randomly generated data sub-sets are split into two groups, i.e., one is called in-bag (IB), while the other is called out-of-the-bag (OOB). To train the model, the IB data will be used while the OOB data is used for model evaluation to resolve to fit (Xing et al., 2019a, b).

$$\text{MSE}_{\text{OOB}} = \sum_{i=1}^{N_{\text{tree}}} \frac{(y - y_i^{\text{OOB}})^2}{N_{\text{tree}}} \quad (1)$$

$$\text{R}^2_{\text{RF}} = 1 - \frac{\text{MSE}_{\text{OOB}}}{\sigma_y^2} \quad (2)$$

Where MSE_{OOB} is the mean square error for out-of-the-bag data. y and y_i^{OOB} represent the predicted and actual targets for OOB data respectively, and N_{tree} shows the number of trees. R^2_{RF} and σ_y^2 shows the determination coefficient and variance of the predicted data, respectively.

Multi Linear Regression (MLR): Linear regression is a statistical tool that is used to identify the linear relationship between the variables and targets. The simple linear regression only deals with the one input variable, when the number of input features is increased then it is called multiple linear regression (MLR).

$$y = \beta_0 + \beta_1 y_1 + \beta_2 y_2 + \dots + \beta_p x_p \quad (3)$$

The term y represents the output of the model, x_i ($i = 1, 2, \dots, p$) represents the number of input features and the term β is the coefficient of the equation. The main goal in MLR is to obtain the optimum coefficient (β) vector value for the given linear regression model (Aladejare et al., 2020; Uyanık & Güler, 2013).

Support Vector Machine (SVM): Support vector machine (SVM) is a supervised machine learning model that deals with classification and regression problems (Y. Bai et al., 2018). The SVM uses the kernel function in both classification and regression problems to develop the linear relationship between input and output data. When the problem is complex and has nonlinear dynamics, the kernel function is particularly useful (McDonald et al., 2017). The kernel function is a function that maps the input features into high dimensional space to linearly separable the variables which cannot be linearly separable in the lower dimension vector.

Decision Tree (DT): A decision tree (DT) algorithm is a supervised machine learning model that predicts the output based on some logical rules. The DT does not depend on the predefined relationship between the input features and the output, unlike other ML methods. By repeatedly splitting the input data set into binary sections, the DT algorithm generates a set of decision rules. The division is examined by the search method for all features until a homogeneous distribution of output is obtained. The best division is called the division for which the output distribution is homogenous. To develop a DT model the hyper-parameter i.e., minimum leaf size should be optimized. The minimum leaf size can be optimized by grid search and Bayesian optimization methods. The DT also has a problem of overfitting, just like other regression algorithms. The process name “pruning” is used in the DT algorithm to prevent overfitting (Mutlu et al., 2019; Ozbas et al., 2019).

3. Results and discussion

In this section, four methods of machine learning (RF, MLR, DT, and SVM) were implemented for the prediction of bio-oil yield using biomass composition and pyrolysis conditions. The recommendations for the practitioners are provided at the end.

3.1. Genetic algorithm for features selection

In machine learning applications, feature or input selection techniques are employed for identifying and eliminating the irrelevant, undesired, and redundant input features or characteristics that do not affect the accuracy of the ML model. However, the process of selecting the most relevant feature is a complex combinatorial task. Two techniques that have been widely used in many studies for feature selection are the Filter method and the Wrapper method (Wah et al., 2018; Zhang et al., 2019). In the filter method, the process of feature selection is carried out as a pre-processing step before machine learning models implementation to the selected features or inputs. The features are selected based on the relationship with the target and between the variables instead of cross-validation performance. This approach may mislead the algorithm for feature selection (Chandrashekar & Sahin, 2014). On the other hand, in the Wrapper methods, the actual ML model is trained on each set of features to select the most relevant features. The limitation of this approach is overfitting due to the training of ML models based on different features (Zhao et al., 2015).

The genetic algorithm (GA) based feature selection is the most advanced and novel approach for feature selection which suggests a natural way for solving such problems (Oliveira et al., 2010). GA is analogous to the Darwinian theory of “Survival of the fittest”. In GA, a set of solutions is selected called population, and a single solution in a set of the population is known as a chromosome. Each chromosome is passed through the fitness function, which is a criterion that estimates the quality of each chromosome. If the desired objective is obtained, the algorithms stop at that point. Otherwise, another set of offspring (a new set of the population) is generated by mutation, crossover, and selection operators (Sharma, 2013). This way, the algorithm is continued until the best result is obtained (termination point is achieved). The main steps of the genetic algorithm are as follows:

Step 1: Randomly generate the set of the population (features).

Step 2: Each feature is passed through the fitness function.

Step 3: If the termination condition is satisfied, move to subsequent steps, otherwise, then the algorithm is initialized again, and a new set of features selected.

Step 4: A new set of features is generated by genetic operations (mutation, crossover, and selection operators)

Step 5: The best feature is selected based on the lowest fitness function value.

Step 6: The best feature is passed through each ML method (Babatunde et al., 2014). Tables 1 and 2 show the GA parameters and selected features for all predictive models, respectively.

Table 1
Parameters used in Genetic Algorithm.

S. No.	Genetic Algorithm Parameters	Value
1	Population Size	50
2	Population Type	bitstring
3	Number of Generations	500
4	Crossover	Crossover arithmetic
5	Crossover Probability	0.8
6	Mutation	Mutation uniform
7	Mutation Probability	0.1
8	Selection Type	Selection roulette
9	Elite Count	2

Table 2
Selected features based on Genetic Algorithm approach.

Model Type	Selected Features	
	Proximate Analysis	Ultimate Analysis
Random Forest	Ash, FC, V, HTT, HR, PS, FR	C, H, N, O, HTT, HR, PS, FR
Multi-linear Regression	Ash, V, HTT, HR, PS, FR	C, H, N, HTT, HR, PS, FR
Support Vector Machine	Ash, V, HR, PS, FR	C, H, HR, FR
Decision Tree	Ash, V, FC, HTT, HR, PS	H, N, HTT, HR

3.2. Optimization of ML methods

The optimal hyper-parameters were obtained for each machine learning model with the 5-fold cross-validation. The input dataset was divided into 80% and 20% for training and testing of the model, respectively. In the case of random forest, the boosted tree model was trained for ultimate and proximate analysis of the biomass. The minimum leaf size, number of trees, and learning rate were optimized. In the case of proximate analysis, the optimum number of trees, learning rate, and leaf size were 98, 0.25, and 16, respectively. While in the case of ultimate analysis, the optimum number of trees, learning rate, and leaf size were 141, 0.5, and 2, respectively. While the grid search methods were used for the optimization of the Decision tree (DT), the fine tree model was trained for both the ultimate and proximate analysis. The optimum minimum leaf size was 1 for the proximate analysis model while the optimum minimum leaf size was 2 for the ultimate analysis. The Gaussian kernel function was used in the case of the SVM model. In the case of ultimate and proximate analysis, the interaction linear model with the quadratic term was chosen for multiple linear regression (MLR).

3.3. Prediction performance of ML methods

After developing and optimizing the machine learning (ML) models as described in Section 3.2, all the ML models including MLR, RF, SVM, and DT were trained using MATLAB Regression Learner Toolbox for evaluating their capability of bio-oil yield prediction using the training dataset which accounts for 80%, and the testing dataset which accounts for 20% of the total 263 data points, respectively. The features selected by the genetic algorithm, as shown in Table 2, were used to train the ML models. The optimized hyper-parameters as discussed in the above-mentioned sections were used for training the ML models. Each ML model was trained for two cases, firstly using the proximate analysis with pyrolysis conditions and secondly, using the ultimate analysis with the pyrolysis conditions. The coefficient of determination (R^2) and root mean square (RMSE) values for the training and testing of the models for both ultimate and proximate analysis cases are presented in Table 3. The comparison of actual and model-predicted yield for both the proximate and ultimate analysis is illustrated in Figs. 4 and 5. The blue solid line in these figures represents the actual data while red dotted circles around the solid line show the predicted values of the proposed ML models.

Table 3
Training and testing performance of machine learning models.

Model Type	Training R^2	Training RMSE	Testing R^2	Testing RMSE
Proximate Analysis				
RF	0.88	2.78	0.98	1.06
MLR	0.66	4.35	0.68	4.20
SVM	0.84	2.78	0.96	1.62
DT	0.77	3.42	0.96	1.69
Ultimate Analysis				
RF	0.92	1.96	0.99	0.56
MLR	0.77	3.47	0.83	2.94
SVM	0.88	2.55	0.96	1.53
DT	0.83	2.99	0.91	2.18

It is evident from Table 3, that the performance of the MLR was not satisfactory in both the proximate and ultimate analysis as compared to that of RF, SVM, and DT. In the case of proximate analysis, the training and testing R^2 scores for the MLR model were 0.66 and 0.677 with RMSE values of 4.35 and 4.20, respectively. While in the case of ultimate analysis, the training and testing R^2 values were 0.77 and 0.83 with RMSE values of 3.47 and 2.94, respectively, which is better than the previously reported MLR models (Tang et al., 2020). The reason for the poor performance of MLR as compared to other ML models shows the non-linear behavior of yield with the variables. However, in the case of proximate analysis, the MLR showed good performance compared to the ultimate analysis. This is because, in the case of ultimate analysis, more features are involved as shown in Table 2. In the case of RF, high performance is achieved as compared to all other models, including MLR, DT, and SVM. This means that the RF is suitable for dealing with the complex correlation between variables and target. Overall, for the proximate analysis, the RF model achieved the acceptable performance with an R^2 value of 0.88 and RMSE value of 2.78 for the training set, while in the case of ultimate analysis, the RF model achieved the highest performance with R^2 and RMSE values of 0.98 and 1.06, respectively. Similarly, for the ultimate analysis, the RF model achieved the best results with R^2 and RMSE of 0.92 and 1.96 for the training set while that of 0.99 and 0.56 for the testing set, respectively. These results are better than the RF models reported in the literature (Tang et al., 2020).

SVM and DT showed low performance in training as compared to RF but achieved good generalization ability by attaining the $R^2 > 0.90$ for a testing dataset for both ultimate and proximate analysis. In the case of proximate analysis, the training R^2 score for the SVM model was 0.84 with an RMSE of 4.35, which was greater than that of the DT model (training R^2 score 0.77, RMSE of 4.35). Both the SVM and DT models showed comparable results (R^2 equals 0.96 each) with a slight difference in RMSE, which was greater than the SVM model ($R^2 = 0.898$ of SVM) are reported earlier (Chen et al., 2018). While in the case of ultimate analysis, the training and testing R^2 values of the SVM model were 0.88 and 0.96 with RMSE values of 2.55 and 1.53, respectively as compared to that of the DT model where training and testing R^2 scores were 0.83 and 0.91, respectively.

In summary, various ML models including MLR, RF, SVM, and DT were established and trained to predict the bio-oil yield. All four ML models have successfully predicted the bio-oil yield based on GA-based selected features. The MLR model showed poor model training and testing performance for both the cases (proximate and ultimate) as compared to the other ML methods including RF, SVM, and DT. On the other hand, the RF model outperformed all the models in training and testing for both cases. In the case of proximate analysis, the performance trends based on R^2 were as follows: for training (RF > SVM > DT > MLR), for testing (RF > SVM > DT > MLR) while in the case of ultimate analysis, the performance trends based on R^2 were as follows: for training (RF > SVM > DT > MLR), for testing (RF > SVM > DT > MLR). Finally, the results of ML models (RF, SVM, and DT) were compared with experimental data taken from the literature, as shown in Table 4. It is evident from the results that proposed ML models can effectively be used for the prediction of bio-oil yield without lab-based experimentations to reduce the time and cost of experimentation.

3.4. Effect of input features on Bio-oil Yield.

The effect of input variables (features selected by Genetic Algorithm) on target (Bio-oil yield) can be shown by the partial dependence analysis. While constructing the partial dependence plots (PDPs), only the influence of selected input features on the target is considered. The PDPs in Fig. 6 depict the influence of selected input features of both the models of RF (Proximate and ultimate) on bio-oil yield.

Fig. 6 (a, b, c, d, e, f, and g) show the partial dependence plots of the proximate model of RF for bio-oil yield prediction. Fig. 6(a) shows the influence of ash content on bio-oil yield. The ash content (2%, 6%),

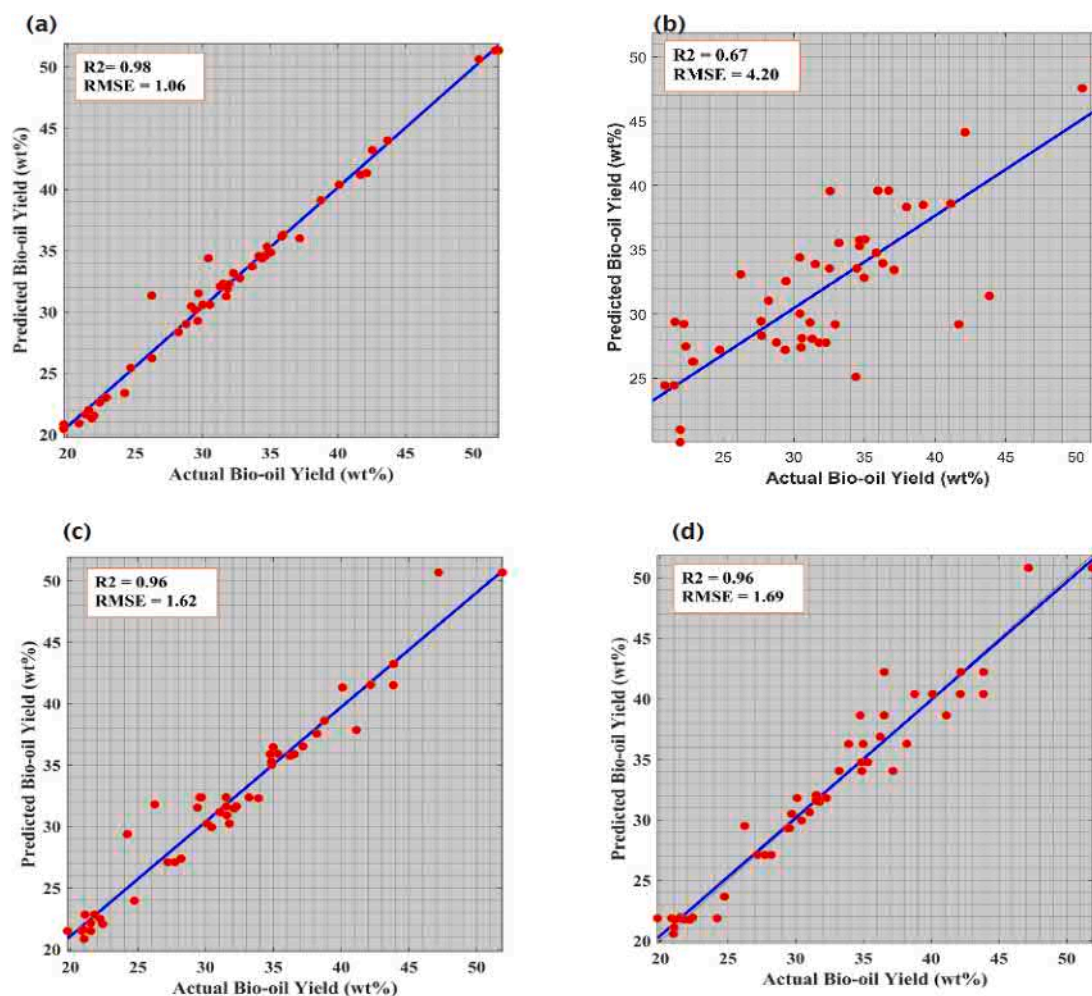


Fig. 4. Comparison of experimental and predicted models output data based on proximate analysis and pyrolysis conditions: (a) Random Forest (b) Multi Linear Regression (c) Support Vector Machine (d) Decision Tree.

showed a positive correlation with the bio-oil yield, while from 6% onwards its value decreased. This trend was comparable with the literature, which concluded that the ash content in the moderate range behaves like a catalyst during the process of pyrolysis due to the presence of metal contents (alkali and alkaline earth metals) (Shadangi & Mohanty, 2014). The presence of alkali metals greatly influences the production of bio-oil yield. In general, the correlation between ash content and bio-oil yield is negative, implying that the higher the ash content, the lower the bio-oil yield. Bio-oil yield decreases with the increase of FC content (12% onwards) as shown in Fig. 6(b). This negative correlation of FC with the bio-oil yield is understandable since increased FC content results in the less volatile matter, which is because of the low ash content in the biomass feedstock. Biomass with higher volatile matter contents, as shown in Fig. 6 (c), would require less space for secondary pyrolysis reactions, resulting in a higher yield of bio-oil, as claimed (Gómez-Monedero et al., 2015). The decrease in the volatile matter from (71%–81%) may appear due to the missing values of samples in this region or the initial presence of the moisture in the biomass feedstock. In Fig. 6(d), the HTT shows the sharp increase from 500 °C onwards, showing a positive correlation with the bio-oil yield. In Fig. 6 (e), the sharp increase in the HR values from 198 °C/min onwards results in higher bio-oil yield values. The higher values of HR mean the availability of the shortest time for the pyrolysis secondary reactions, hence resulting in a higher bio-oil yield. Furthermore, it can be seen from Fig. 6(f) that PS from 0.6 mm onwards the bio-oil yield increases with an increase in PS, and a negative correlation was seen for other

values of PS, showing the complex relationship of PS with the bio-oil yield due to heat and mass transfer limitation. FR showed a positive correlation with the bio-oil yield, because of the reduction of secondary reactions of pyrolysis (volatile secondary decomposition) as shown in Fig. 6(g).

Fig. 6 (a', b', c', and d') shows the partial dependence plots of the RF Ultimate model for the bio-oil yield prediction. Fig. 6(a') and 6(b') show the correlation of carbon and hydrogen with bio-oil yield, respectively. Figures illustrate that bio-oil yield increases with increasing both the amount of carbon and hydrogen, as reported by other researchers (Chouhan & Sarma, 2013). Moreover, by increasing the amount of carbon and hydrogen, the calorific value of the bio-oil increases. Fig. 6 (c') shows the influence of nitrogen content on the bio-oil yield. The direct correlation between nitrogen and bio-oil yield can be seen. Fig. 6 (d') shows the influence of oxygen content on the bio-oil yield. The presence of oxygen content in the biomass reduces the bio-oil yield (Chouhan & Sarma, 2013), hence shows a negative correlation with the bio-oil yield since the presence of oxygen content in the bio-oil degrades its quality (Chen et al., 2019). Fig. 6(e') and 6(f') show the correlation of HTT and HR with the bio-oil yield, respectively. The increasing trend in both the plots shows that both the HTT and HR have a direct influence on the bio-oil yield, for the same reasons as mentioned earlier in Fig. 6 (d) and 6(e). As mentioned earlier in Fig. 6(f), the particle size showed a complex and non-uniform correlation with the bio-oil yield and shows a similar trend in the case of Fig. 6(g') as well. Similarly, from Fig. 6(h'), FR showed a positive correlation with the bio-oil yield, because of the

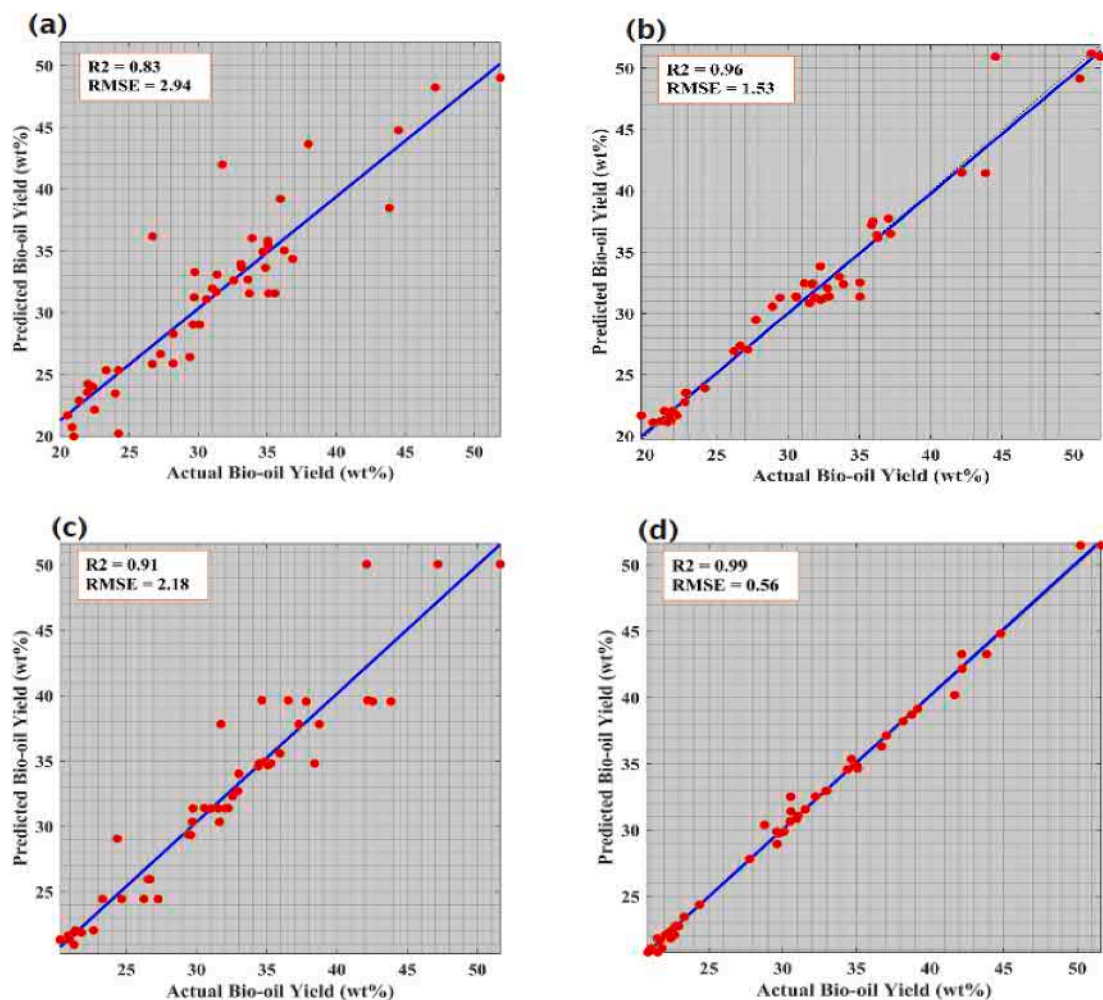


Fig. 5. Comparison of experimental and predicted models output data based on ultimate analysis and pyrolysis conditions: (a) Multi Linear Regression (b) Support Vector Machine (c) Decision Tree (d) Random Forest.

Table 4

Comparison of bio-oil yield predicted by RF, SVM, and DT with experimental values.

RF (ultimate analysis)	RF (proximate analysis)	SVM (ultimate analysis)	SVM (proximate analysis)	DT (ultimate analysis)	DT (proximate analysis)	Experimental Bio-oil yield	References
43.80	35.89	35.68	38.82	39.64	31.80	47.73	(Abnisa et al., 2013)
45.73	45.06	43.54	42.109	39.64	45.75	45.75	(Abnisa et al., 2013)
34.22	34.18	33.96	34.02	39.64	34.62	26.5	(Chen et al., 2015)
34.22	34.18	33.96	34.02	39.64	34.62	35.2	(Chen et al., 2015)
31.02	30.42	31.74	31.17	30.14	29.49	31.05	(Pütün et al., 2001)
31.02	30.42	31.74	31.17	30.14	29.49	31.05	(Pütün et al., 2001)
29.41	30.57	30.83	31.64	29.39	29.30	29.80	(Bordoloi et al., 2016)
29.41	30.57	30.83	31.64	29.39	29.30	33.24	(Bordoloi et al., 2016)
30.51	28.13	30.68	31.70	31.38	31.59	25.58	(Demiral and Ashi Ayan, 2011)
28.34	26.62	30.91	29.43	29.06	23.65	38.97	(Garg et al., 2016)

reduction of secondary reactions of pyrolysis (volatile secondary decomposition).

3.5. Recommendations for practitioners

Based on results and discussion, the following recommendations are presented to practitioners:

1. MLR model should not be used for the bio-oil yield prediction due to its poor generalization performance, which shows the non-linear behavior of yield with the variables.
2. RF outperformed other ML models (MLR, DT, and SVM) with high generalization performance and is highly recommended when dealing with the complex correlation between variables and target.
3. The SVM and DT have comparable results with high generalization capabilities for both the cases of ultimate and proximate analysis.
4. No study has reported the application of deep learning predictive models for bio-oil yield prediction, which should be incorporated in future studies.
5. The dataset for ML models can be improved by including the effect of biomass components (cellulose, hemicellulose, and lignin) and pyrolysis reactor volume and design parameters on bio-oil yield.

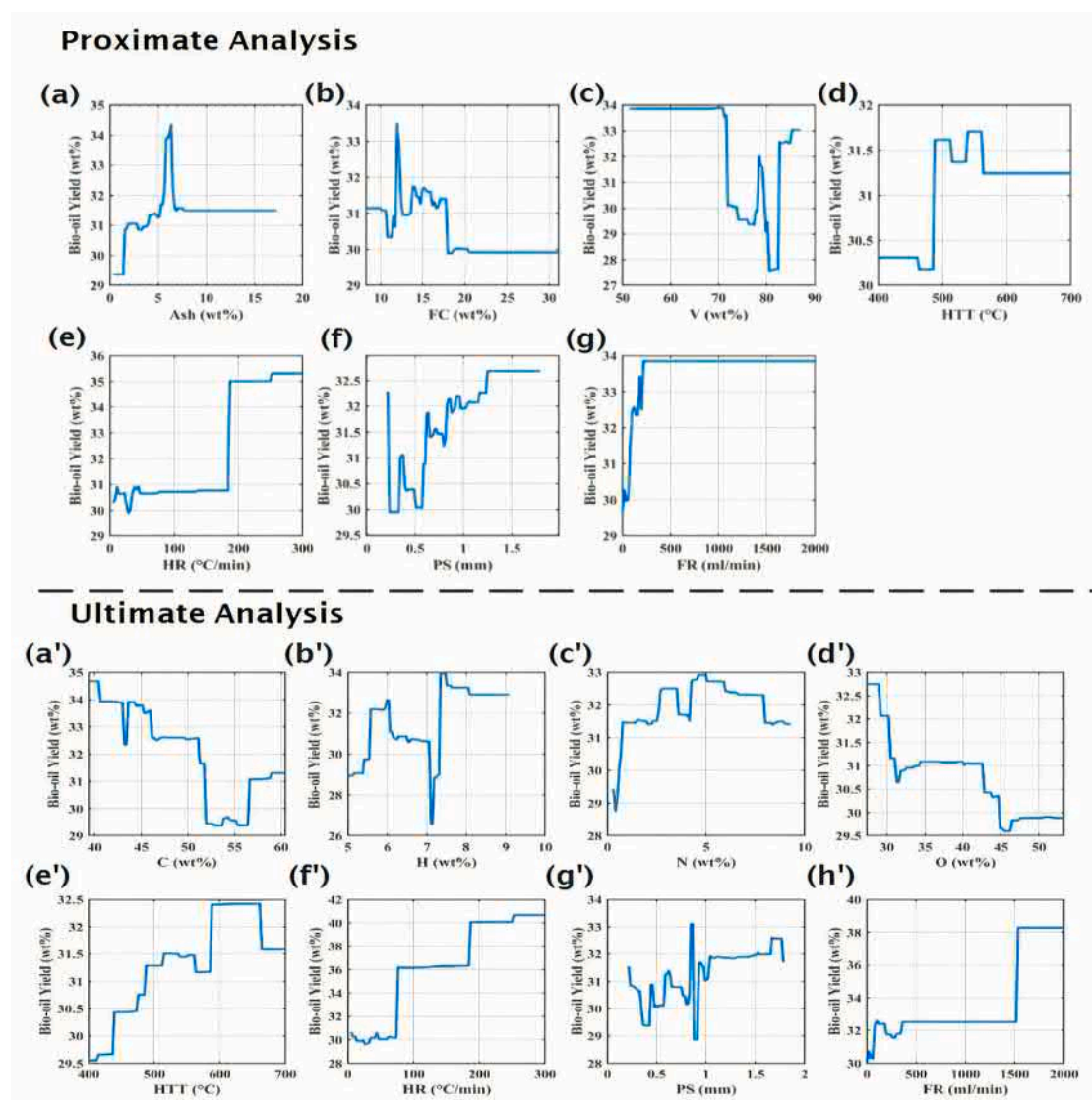


Fig. 6. Partial Differential Plots for Bio-oil yield in the case proximate and ultimate analysis.

4. Conclusion

The bio-oil yield of biomass was successfully predicted with four types of machine learning methods using biomass composition and pyrolysis conditions. The random forest model showed better prediction performance ($R^2 \sim 0.98$ and $RMSE \sim 0.81$) as compared to SVM ($R^2 \sim 0.96$), DT ($R^2 \sim 0.93$), and MLR ($R^2 \sim 0.75$). Furthermore, the RF, SVM, and DT models were validated with the experimental studies in the literature. PDP analysis showed that the optimal parameters selected using a genetic algorithm-based approach have a great influence on bio-oil yield. This research offered an operational methodology for perspectives in-depth studies on biomass conversion.

CCRediT authorship contribution statement

Zahid Ullah: Investigation, Writing - original draft. **Muzammil Khan:** Formal analysis, Writing - original draft. **Salman Raza Naqvi:** Methodology, Supervision, Funding acquisition, Writing - review & editing. **Wasif Farooq:** Validation, Writing - review & editing. **Haiping Yang:** Formal analysis, Validation. **Shurong Wang:** Resources, Validation. **Dai-Viet N. Vo:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2021.125292>.

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